

latex-math Macros

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Latex macros like `\frac{#1}{#2}` with arguments are displayed as $\frac{\#1}{\#2}$.

Note that macro declarations may only span a single line to be displayed correctly in the below tables.

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basic-math

Macro	Notation	Comment
<code>\N</code>	$\mathbb{N}$	N, naturals
<code>\Z</code>	$\mathbb{Z}$	Z, integers
<code>\Q</code>	$\mathbb{Q}$	Q, rationals
<code>\R</code>	$\mathbb{R}$	R, reals
<code>\C</code>	$\mathbb{C}$	C, complex
<code>\continuous</code>	$\mathcal{C}$	C, space of continuous functions
<code>\M</code>	$\mathcal{M}$	machine numbers
<code>\epsm</code>	ϵ_m	maximum error
<code>\setzo</code>	$\{0, 1\}$	set 0, 1
<code>\setmp</code>	$\{-1, +1\}$	set -1, 1
<code>\unitint</code>	$[0, 1]$	unit interval
<code>\xt</code>	$\tilde{x}$	x tilde
<code>\argmin</code>	arg min	argmin
<code>\argmax</code>	arg max	argmax
<code>\argminlim</code>	arg min	argmin with limits
<code>\argmaxlim</code>	arg max	argmax with limits
<code>\sign</code>	sign	sign, signum
<code>\I</code>	$\mathbb{I}$	I, indicator
<code>\order</code>	$\mathcal{O}$	O, order
<code>\bigO</code>	$\mathcal{O}$	Big-O Landau
<code>\littleo</code>	o	Little-o Landau
<code>\pd</code>	$\frac{\partial \#1}{\partial \#2}$	partial derivative
<code>\floorlr</code>	$\lfloor \#1 \rfloor$	floor
<code>\ceillr</code>	$\lceil \#1 \rceil$	ceiling
<code>\indep</code>	$\perp\!\!\!\perp$	independence symbol
<code>\xv</code>	$\boldsymbol{x}$	vector x (bold)
<code>\xtil</code>	$\tilde{\boldsymbol{x}}$	vector x-tilde (bold)
<code>\yv</code>	$\boldsymbol{y}$	vector y (bold)
<code>\vv</code>	$\boldsymbol{v}$	vector v (bold)
<code>\wv</code>	$\boldsymbol{w}$	vector w (bold)
<code>\uv</code>	$\boldsymbol{u}$	vector u (bold)
<code>\muv</code>	$\boldsymbol{\mu}$	vector mu (bold)
<code>\nuv</code>	$\boldsymbol{\nu}$	vector nu (bold)
<code>\sumin</code>	$\sum_{i=1}^n$	summation from i=1 to n
<code>\sumim</code>	$\sum_{i=1}^m$	summation from i=1 to m
<code>\sumjn</code>	$\sum_{j=1}^n$	summation from j=1 to p
<code>\sumjp</code>	$\sum_{j=1}^p$	summation from j=1 to p

<code>\sumik</code>	$\sum_{i=1}^k$	summation from i=1 to k
<code>\sumkg</code>	$\sum_{k=1}^g$	summation from k=1 to g
<code>\sumjg</code>	$\sum_{j=1}^g$	summation from j=1 to g
<code>\summM</code>	$\sum_{m=1}^M$	summation from m=1 to M
<code>\meanin</code>	$\frac{1}{n} \sum_{i=1}^n$	mean from i=1 to n
<code>\meanim</code>	$\frac{1}{m} \sum_{i=1}^m$	mean from i=1 to n
<code>\meankg</code>	$\frac{1}{g} \sum_{k=1}^g$	mean from k=1 to g
<code>\meanmM</code>	$\frac{1}{M} \sum_{m=1}^M$	mean from m=1 to M
<code>\prodin</code>	$\prod_{i=1}^n$	product from i=1 to n
<code>\prodkg</code>	$\prod_{k=1}^g$	product from k=1 to g
<code>\prodjg</code>	$\prod_{j=1}^p$	product from j=1 to p
<code>\one</code>	1	1, unitvector
<code>\zero</code>	0	0-vector
<code>\id</code>	I	I, identity
<code>\diag</code>	diag	diag, diagonal
<code>\trace</code>	tr	tr, trace
<code>\spn</code>	span	span
<code>\scp</code>	$\langle \#1, \#2 \rangle$	$\langle ., . \rangle$, scalarproduct
<code>\mat</code>	$(\#1)$	short pmatrix command
<code>\Amat</code>	A	matrix A
<code>\Deltab</code>	Δ	error term for vectors
<code>\P</code>	P	P, probability
<code>\E</code>	E	E, expectation
<code>\var</code>	Var	Var, variance
<code>\cov</code>	Cov	Cov, covariance
<code>\corr</code>	Corr	Corr, correlation
<code>\normal</code>	$\mathcal{N}$	N of the normal distribution
<code>\iid</code>	$i.i.d$	dist with i.i.d superscript
<code>\distas</code>	$\#^1$	... is distributed as ...
<code>\fx</code>	$f(\boldsymbol{x})$	f(x), continuous prediction function

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basic-ml

Macro	Notation	Comment
<code>\Xspace</code>	$\mathcal{X}$	X, input space
<code>\Yspace</code>	$\mathcal{Y}$	Y, output space
<code>\Zspace</code>	$\mathcal{Z}$	Z, space of sampled datapoints
<code>\nset</code>	$\{1, \dots, n\}$	set from 1 to n
<code>\pset</code>	$\{1, \dots, p\}$	set from 1 to p
<code>\gset</code>	$\{1, \dots, g\}$	set from 1 to g
<code>\Pxy</code>	$\mathbb{P}_{xy}$	P_{xy}
<code>\Exy</code>	$\mathbb{E}_{xy}$	E_{xy} : Expectation over random variables xy
<code>\xy</code>	$(\mathbf{x}, y)$	observation (x, y)
<code>\xvec</code>	$(x_1, \dots, x_p)^\top$	(x1, ..., xp)
<code>\Xmat</code>	$\mathbf{X}$	Design matrix
<code>\allDatasets</code>	$\mathcal{D}$	The set of all datasets
<code>\allDatasetsn</code>	$\mathcal{D}_n$	The set of all datasets of size n
<code>\D</code>	$\mathcal{D}$	D, data
<code>\Dn</code>	$\mathcal{D}_n$	D_n , data of size n
<code>\Dtrain</code>	$\mathcal{D}_{\text{train}}$	D_{train} , training set
<code>\Dtest</code>	$\mathcal{D}_{\text{test}}$	D_{test} , test set
<code>\xyi</code>	$(\mathbf{x}^{(\#1)}, y^{(\#1)})$	(x~i, y~i), i-th observation
<code>\Dset</code>	$((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$	{(x1,y1), ..., (xn,yn)}, data
<code>\defAllDatasetsn</code>	$(\mathcal{X} \times \mathcal{Y})^n$	Def. of the set of all datasets of size n
<code>\defAllDatasets</code>	$\bigcup_{n \in \mathbb{N}} (\mathcal{X} \times \mathcal{Y})^n$	Def. of the set of all datasets
<code>\xdat</code>	$\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$	{x1, ..., xn}, input data
<code>\ydat</code>	$\{\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(n)}\}$	{y1, ..., yn}, input data
<code>\yvec</code>	$(y^{(1)}, \dots, y^{(n)})^\top$	(y1, ..., yn), vector of outcomes
<code>\greekxi</code>	$\mathbf{x}^{(i)}$	Greek letter xi
<code>\xi</code>	$\mathbf{x}^{(\#1)}$	x~i, i-th observed value of x
<code>\yi</code>	$y^{(\#1)}$	y~i, i-th observed value of y
<code>\xivec</code>	$(x_1^{(i)}, \dots, x_p^{(i)})^\top$	(x1~i, ..., xp~i), i-th observation vector
<code>\xj</code>	$\mathbf{x}_j$	x~j, j-th feature
<code>\xjvec</code>	$(x_j^{(1)}, \dots, x_j^{(n)})^\top$	(x1~j, ..., xn~j), j-th feature vector
<code>\phiv</code>	ϕ	Basis transformation function phi
<code>\phixi</code>	$\phi^{(i)}$	Basis transformation of xi: phi~i := phi(xi)
<code>\lamv</code>	$\boldsymbol{\lambda}$	lambda vector, hyperconfiguration vector
<code>\Lam</code>	Λ	Lambda, space of all hpos
<code>\preimageInducer</code>	$(\bigcup_{n \in \mathbb{N}} (\mathcal{X} \times \mathcal{Y})^n) \times \Lambda$	Set of all datasets times the hyperparameter space
<code>\preimageInducerShort</code>	$\mathcal{D} \times \Lambda$	Set of all datasets times the hyperparameter space
<code>\ind</code>	$\mathcal{I}$	Inducer, inducing algorithm, learning algorithm
<code>\ftrue</code>	f_{true}	True underlying function (if a statistical model is assumed)
<code>\ftruex</code>	$f_{\text{true}}(\mathbf{x})$	True underlying function (if a statistical model is assumed)
<code>\fdomains</code>	$f : \mathcal{X} \rightarrow \mathbb{R}^g$	f with domain and co-domain

<code>\Hspace</code>	$\mathcal{H}$	hypothesis space where f is from
<code>\Hall</code>	$\mathcal{H}_{\text{all}}$	unrestricted hypothesis space
<code>\fbayes</code>	f^*	Bayes-optimal model
<code>\fxbayes</code>	$f^*(\mathbf{x})$	Bayes-optimal model
<code>\fkx</code>	$f_{\#1}(\mathbf{x})$	$f_{\sim j}(\mathbf{x})$, discriminant component function
<code>\fhspace</code>	$\hat{f}_{\mathcal{H}}$	fhat_H
<code>\fh</code>	$\hat{f}$	f hat, estimated prediction function
<code>\fxh</code>	$\hat{f}(\mathbf{x})$	$\text{fhat}(\mathbf{x})$
<code>\fxt</code>	$f(\mathbf{x} \mid \boldsymbol{\theta})$	$f(\mathbf{x} \mid \text{theta})$
<code>\fxi</code>	$f(\mathbf{x}^{(i)})$	$f(\mathbf{x}^{\sim(i)})$
<code>\fxih</code>	$\hat{f}(\mathbf{x}^{(i)})$	$f(\mathbf{x}^{\sim(i)})$
<code>\fxit</code>	$f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta})$	$f(\mathbf{x}^{\sim(i)} \mid \text{theta})$
<code>\fhD</code>	$\hat{f}_{\mathcal{D}}$	fhat_D , estimate of f based on D
<code>\fhDtrain</code>	$\hat{f}_{\mathcal{D}_{\text{train}}}$	fhat_Dtrain , estimate of f based on D
<code>\fhDnlam</code>	$\hat{f}_{\mathcal{D}_n, \lambda}$	model learned on Dn with hp lambda
<code>\fhDlam</code>	$\hat{f}_{\mathcal{D}, \lambda}$	model learned on D with hp lambda
<code>\fhDnlams</code>	$\hat{f}_{\mathcal{D}_n, \lambda^*}$	model learned on Dn with optimal hp lambda
<code>\fhDlams</code>	$\hat{f}_{\mathcal{D}, \lambda^*}$	model learned on D with optimal hp lambda
<code>\hx</code>	$h(\mathbf{x})$	$h(\mathbf{x})$, discrete prediction function
<code>\hh</code>	$\hat{h}$	h hat
<code>\hxx</code>	$\hat{h}(\mathbf{x})$	$\text{hhat}(\mathbf{x})$
<code>\hxt</code>	$h(\mathbf{x} \mid \boldsymbol{\theta})$	$h(\mathbf{x} \mid \text{theta})$
<code>\hxi</code>	$h(\mathbf{x}^{(i)})$	$h(\mathbf{x}^{\sim(i)})$
<code>\hxit</code>	$h(\mathbf{x}^{(i)} \mid \boldsymbol{\theta})$	$h(\mathbf{x}^{\sim(i)} \mid \text{theta})$
<code>\hbayes</code>	h^*	Bayes-optimal classification model
<code>\hxbayes</code>	$h^*(\mathbf{x})$	Bayes-optimal classification model
<code>\yh</code>	$\hat{y}$	yhat for prediction of target
<code>\yih</code>	$\hat{y}^{(i)}$	$\text{yhat}^{\sim(i)}$ for prediction of ith targiet
<code>\resi</code>	$y^{(i)} - \hat{y}^{(i)}$	
<code>\thetah</code>	$\hat{\boldsymbol{\theta}}$	theta hat
<code>\thetav</code>	$\boldsymbol{\theta}$	theta vector
<code>\thetavh</code>	$\hat{\boldsymbol{\theta}}$	theta vector hat
<code>\thetat</code>	$\boldsymbol{\theta}^{[\#1]}$	$\text{theta}^{\sim[t]}$ in optimization
<code>\thetatn</code>	$\boldsymbol{\theta}^{[\#1+1]}$	$\text{theta}^{\sim[t+1]}$ in optimization
<code>\thetahDnlam</code>	$\hat{\boldsymbol{\theta}}_{\mathcal{D}_n, \lambda}$	theta learned on Dn with hp lambda
<code>\thetahDlam</code>	$\hat{\boldsymbol{\theta}}_{\mathcal{D}, \lambda}$	theta learned on D with hp lambda
<code>\mint</code>	$\min_{\boldsymbol{\theta} \in \Theta}$	min problem theta
<code>\argmint</code>	$\arg \min_{\boldsymbol{\theta} \in \Theta}$	argmin theta
<code>\pdf</code>	p	p
<code>\pdfx</code>	$p(\mathbf{x})$	$p(\mathbf{x})$
<code>\pixt</code>	$\pi(\mathbf{x} \mid \boldsymbol{\theta})$	$\text{pi}(\mathbf{x} \mid \text{theta})$, pdf of x given theta
<code>\pixit</code>	$\pi(\mathbf{x}^{(\#1)} \mid \boldsymbol{\theta})$	$\text{pi}(\mathbf{x}^{\sim i} \mid \text{theta})$, pdf of x given theta
<code>\pixii</code>	$\pi(\mathbf{x}^{(\#1)})$	$\text{pi}(\mathbf{x}^{\sim i})$, pdf of i-th x
<code>\pdfxy</code>	$p(\mathbf{x}, y)$	$p(\mathbf{x}, y)$

<code>\pdfxyt</code>	$p(\mathbf{x}, y \mid \boldsymbol{\theta})$	$p(\mathbf{x}, y \mid \text{theta})$
<code>\pdfxyit</code>	$p(\mathbf{x}^{(i)}, y^{(i)} \mid \boldsymbol{\theta})$	$p(\mathbf{x}^{\wedge(i)}, y^{\wedge(i)} \mid \text{theta})$
<code>\pdfxyk</code>	$p(\mathbf{x} \mid y = \#1)$	$p(\mathbf{x} \mid y = k)$
<code>\lpdfxyk</code>	$\log p(\mathbf{x} \mid y = \#1)$	$\log p(\mathbf{x} \mid y = k)$
<code>\pdfxiyk</code>	$p(\mathbf{x}^{(i)} \mid y = \#1)$	$p(\mathbf{x}^{\wedge i} \mid y = k)$
<code>\pik</code>	$\pi_{\#1}$	pi_k, prior
<code>\pih</code>	$\hat{\pi}$	$\text{pi hat, estimated prior (binary classification)}$
<code>\pikh</code>	$\hat{\pi}_{\#1}$	$\text{pi_k hat, estimated prior}$
<code>\lpik</code>	$\log \pi_{\#1}$	$\log \text{pi_k, log of the prior}$
<code>\pit</code>	$\pi(\boldsymbol{\theta})$	$\text{Prior probability of parameter theta}$
<code>\post</code>	$\mathbb{P}(y = 1 \mid \mathbf{x})$	$P(y = 1 \mid \mathbf{x}), \text{ post. prob for } y=1$
<code>\postk</code>	$\mathbb{P}(y = \#1 \mid \mathbf{x})$	$P(y = k \mid \mathbf{x}), \text{ post. prob for } y=k$
<code>\pidomains</code>	$\pi : \mathcal{X} \rightarrow [0, 1]$	$\text{pi with domain and co-domain}$
<code>\pibayes</code>	π^*	$\text{Bayes-optimal classification model}$
<code>\pixbayes</code>	$\pi^*(\mathbf{x})$	$\text{Bayes-optimal classification model}$
<code>\piastxttil</code>	$\pi^*(\tilde{\mathbf{x}})$	$\text{Bayes-optimal classification model}$
<code>\piastkxttil</code>	$\pi_k^*(\tilde{\mathbf{x}})$	$\text{Bayes-optimal classification model for k-th class}$
<code>\pix</code>	$\pi(\mathbf{x})$	$\text{pi}(\mathbf{x}), P(y = 1 \mid \mathbf{x})$
<code>\piv</code>	$\boldsymbol{\pi}$	$\text{pi, bold, as vector}$
<code>\pikx</code>	$\pi_{\#1}(\mathbf{x})$	$\text{pi_k}(\mathbf{x}), P(y = k \mid \mathbf{x})$
<code>\pikxt</code>	$\pi_{\#1}(\mathbf{x} \mid \boldsymbol{\theta})$	$\text{pi_k}(\mathbf{x} \mid \text{theta}), P(y = k \mid \mathbf{x}, \text{theta})$
<code>\pixh</code>	$\hat{\pi}(\mathbf{x})$	$\text{pi}(\mathbf{x}) \text{ hat}, P(y = 1 \mid \mathbf{x}) \text{ hat}$
<code>\pikxh</code>	$\hat{\pi}_{\#1}(\mathbf{x})$	$\text{pi_k}(\mathbf{x}) \text{ hat}, P(y = k \mid \mathbf{x}) \text{ hat}$
<code>\pixih</code>	$\hat{\pi}(\mathbf{x}^{(i)})$	$\text{pi}(\mathbf{x}^{\wedge(i)}) \text{ with hat}$
<code>\pikxih</code>	$\hat{\pi}_{\#1}(\mathbf{x}^{(i)})$	$\text{pi_k}(\mathbf{x}^{\wedge(i)}) \text{ with hat}$
<code>\pdfygxt</code>	$p(y \mid \mathbf{x}, \boldsymbol{\theta})$	$p(y \mid \mathbf{x}, \text{theta})$
<code>\pdfyigxit</code>	$p(y^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\theta})$	$p(y^{\wedge i} \mid \mathbf{x}^{\wedge i}, \text{theta})$
<code>\lpdfygxt</code>	$\log p(y \mid \mathbf{x}, \boldsymbol{\theta})$	$\log p(y \mid \mathbf{x}, \text{theta})$
<code>\lpdfyigxit</code>	$\log p(y^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\theta})$	$\log p(y^{\wedge i} \mid \mathbf{x}^{\wedge i}, \text{theta})$
<code>\bayesrulek</code>	$\frac{\mathbb{P}(\mathbf{x} \mid y=\#1)\mathbb{P}(y=\#1)}{\mathbb{P}(\mathbf{x})}$	Bayes rule
<code>\muk</code>	$\boldsymbol{\mu}_{\#1}$	$\text{mean vector of class-k Gaussian (discr analysis)}$
<code>\mukh</code>	$\hat{\boldsymbol{\mu}}_{\#1}$	$\text{estimated mean vector of class-k Gaussian (discr analysis)}$
<code>\rx</code>	$r(\mathbf{x})$	residual
<code>\eps</code>	ϵ	$\text{residual, stochastic}$
<code>\epsv</code>	$\boldsymbol{\epsilon}$	$\text{residual, stochastic, as vector}$
<code>\epsi</code>	$\epsilon^{(i)}$	$\text{epsilon}^{\wedge i}, \text{residual, stochastic}$
<code>\epsh</code>	$\hat{\epsilon}$	$\text{residual, estimated}$
<code>\epsvh</code>	$\hat{\boldsymbol{\epsilon}}$	$\text{residual, estimated, vector}$
<code>\yf</code>	$y f(\mathbf{x})$	$y f(\mathbf{x}), \text{ margin}$
<code>\yfi</code>	$y^{(i)} f(\mathbf{x}^{(i)})$	$y^{\wedge i} f(\mathbf{x}^{\wedge i}), \text{ margin}$
<code>\Sigmah</code>	$\hat{\Sigma}$	$\text{estimated covariance matrix}$
<code>\Sigmahj</code>	$\hat{\Sigma}_j$	$\text{estimated covariance matrix for the j-th class}$
<code>\nux</code>	$\nu(\mathbf{x})$	$\text{nu}(\mathbf{x}) = y * f(\mathbf{x})$
<code>\Lyf</code>	$L(y, f)$	$L(y, f), \text{ loss function}$
<code>\Lypi</code>	$L(y, \pi)$	$L(y, \text{pi}), \text{ loss function}$

<code>\Lxy</code>	$L(y, f(\mathbf{x}))$	$L(y, f(\mathbf{x}))$, loss function
<code>\Lxyi</code>	$L(y^{(i)}, f(\mathbf{x}^{(i)}))$	loss of observation
<code>\Lxyt</code>	$L(y, f(\mathbf{x} \boldsymbol{\theta}))$	loss with f parameterized
<code>\Lxyit</code>	$L(y^{(i)}, f(\mathbf{x}^{(i)} \boldsymbol{\theta}))$	loss of observation with f parameterized
<code>\Lxym</code>	$L(y^{(i)}, f(\tilde{\mathbf{x}}^{(i)} \boldsymbol{\theta}))$	loss of observation with f parameterized
<code>\Lpixy</code>	$L(y, \pi(\mathbf{x}))$	loss in classification
<code>\Lpiy</code>	$L(y, \pi)$	loss in classification
<code>\Lpiv</code>	$L(y, \boldsymbol{\pi})$	loss in classification
<code>\Lpixyi</code>	$L(y^{(i)}, \pi(\mathbf{x}^{(i)}))$	loss of observation in classification
<code>\Lpixyt</code>	$L(y, \pi(\mathbf{x} \boldsymbol{\theta}))$	loss with pi parameterized
<code>\Lpixyit</code>	$L(y^{(i)}, \pi(\mathbf{x}^{(i)} \boldsymbol{\theta}))$	loss of observation with pi parameterized
<code>\Lhy</code>	$L(y, h)$	$L(y, h)$, loss function on discrete classes
<code>\Lhxy</code>	$L(y, h(\mathbf{x}))$	$L(y, h(\mathbf{x}))$, loss function on discrete classes
<code>\Lr</code>	$L(r)$	$L(r)$, loss defined on residual (reg) / margin (classif)
<code>\lone</code>	$ y - f(\mathbf{x}) $	L1 loss
<code>\ltwo</code>	$(y - f(\mathbf{x}))^2$	L2 loss
<code>\lbernoullimp</code>	$\ln(1 + \exp(-y \cdot f(\mathbf{x})))$	Bernoulli loss for -1, +1 encoding
<code>\lbernoullizo</code>	$-y \cdot f(\mathbf{x}) + \log(1 + \exp(f(\mathbf{x})))$	Bernoulli loss for 0, 1 encoding
<code>\lcrossent</code>	$-y \log(\pi(\mathbf{x})) - (1 - y) \log(1 - \pi(\mathbf{x}))$	cross-entropy loss
<code>\lbrier</code>	$(\pi(\mathbf{x}) - y)^2$	Brier score
<code>\risk</code>	$\mathcal{R}$	$\mathcal{R}$, risk
<code>\riskbayes</code>	$\mathcal{R}^*$	
<code>\riskf</code>	$\mathcal{R}(f)$	$\mathcal{R}(f)$, risk
<code>\riskdef</code>	$\mathbb{E}_{y \mathbf{x}}(L(y, f(\mathbf{x})))$	risk def (expected loss)
<code>\riskt</code>	$\mathcal{R}(\boldsymbol{\theta})$	$\mathcal{R}(\boldsymbol{\theta})$, risk
<code>\riske</code>	$\mathcal{R}_{\text{emp}}$	$\mathcal{R}_{\text{emp}}$, empirical risk w/o factor 1 / n
<code>\riskeb</code>	$\bar{\mathcal{R}}_{\text{emp}}$	$\bar{\mathcal{R}}_{\text{emp}}$, empirical risk w/ factor 1 / n
<code>\riskef</code>	$\mathcal{R}_{\text{emp}}(f)$	$\mathcal{R}_{\text{emp}}(f)$
<code>\risket</code>	$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$	$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
<code>\riskr</code>	$\mathcal{R}_{\text{reg}}$	$\mathcal{R}_{\text{reg}}$, regularized risk
<code>\riskrt</code>	$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$	$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$
<code>\riskrf</code>	$\mathcal{R}_{\text{reg}}(f)$	$\mathcal{R}_{\text{reg}}(f)$
<code>\riskrth</code>	$\hat{\mathcal{R}}_{\text{reg}}(\boldsymbol{\theta})$	hat $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$
<code>\risketh</code>	$\hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\theta})$	hat $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
<code>\LL</code>	$\mathcal{L}$	$\mathcal{L}$, likelihood
<code>\LLt</code>	$\mathcal{L}(\boldsymbol{\theta})$	$\mathcal{L}(\boldsymbol{\theta})$, likelihood
<code>\LLtx</code>	$\mathcal{L}(\boldsymbol{\theta} \mathbf{x})$	$\mathcal{L}(\boldsymbol{\theta} \mathbf{x})$, likelihood
<code>\logl</code>	ℓ	ℓ , log-likelihood
<code>\loglt</code>	$\ell(\boldsymbol{\theta})$	$\ell(\boldsymbol{\theta})$, log-likelihood
<code>\logltx</code>	$\ell(\boldsymbol{\theta} \mathbf{x})$	$\ell(\boldsymbol{\theta} \mathbf{x})$, log-likelihood
<code>\errtrain</code>	$\text{err}_{\text{train}}$	training error
<code>\errtest</code>	err_{test}	test error
<code>\errexpr</code>	$\overline{\text{err}_{\text{test}}}$	avg training error
<code>\thx</code>	$\boldsymbol{\theta}^\top \mathbf{x}$	linear model
<code>\olsest</code>	$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$	OLS estimator in LM

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ml-ensembles

Macro	Notation	Comment
\bl	$b^{[\#1]}$	baselearner, default m
\blh	$\hat{b}^{[\#1]}$	estimated base learner, default m
\blx	$b^{[\#1]}(\mathbf{x})$	baselearner, default m
\blf	$f^{[\#1]}$	baselearner: scores, default m
\blfh	$\hat{f}^{[\#1]}$	estimated baselearner: scores, default m
\blfhx	$\hat{f}^{[\#1]}(\mathbf{x})$	estimated baselearner: scores of x, default m
\bll	$h^{[\#1]}$	baselearner: hard labels, default m
\bllh	$\hat{h}^{[\#1]}$	estimated baselearner: hard labels, default m
\bllhx	$\hat{h}^{[\#1]}(\mathbf{x})$	estimated baselearner: hard labels of x, default m
\blp	$\pi^{[\#1]}$	baselearner: probabilities, default m
\blph	$\hat{\pi}^{[\#1]}$	estimated baselearner: probabilities, default m
\blphxk	$\hat{\pi}_k^{[\#1]}(\mathbf{x})$	estimated baselearner: probabilities of x for class k, default m
\fM	$f^{[M]}(\mathbf{x})$	ensembled predictor
\fMh	$\hat{f}^{[M]}(\mathbf{x})$	estimated ensembled predictor
\ambifM	$\Delta(f^{[M]}(\mathbf{x}))$	ambiguity/instability of ensemble
\betam	$\beta^{[\#1]}$	weight of basemodel m
\betamh	$\hat{\beta}^{[\#1]}$	weight of basemodel m with hat
\betaM	$\beta^{[M]}$	last baselearner
\ib	IB	In-Bag (IB)
\ibm	$\text{IB}^{[m]}$	In-Bag (IB) for m-th bootstrap
\oob	OOB	Out-of-Bag (OOB)
\oobm	$\text{OOB}^{[m]}$	Out-of-Bag (OOB) for m-th bootstrap
\fm	$f^{[\#1]}$	prediction in iteration m
\fmh	$\hat{f}^{[\#1]}$	prediction in iteration m
\fmd	$f^{[\#1-1]}$	prediction m-1
\fmdh	$\hat{f}^{[\#1-1]}$	prediction m-1
\errm	$\text{err}^{[\#1]}$	weighted in-sample misclassification rate
\wm	$w^{[\#1]}$	weight vector of basemodel m
\wmi	$w^{[\#1](i)}$	weight of obs i of basemodel m
\thetam	$\boldsymbol{\theta}^{[\#1]}$	parameters of basemodel m
\thetamh	$\hat{\boldsymbol{\theta}}^{[\#1]}$	parameters of basemodel m with hat
\blxt	$b(\mathbf{x}, \boldsymbol{\theta}^{[\#1]})$	baselearner, default m
\ens	$\sum_{m=1}^M \beta^{[m]} b(\mathbf{x}, \boldsymbol{\theta}^{[m]})$	ensemble
\rmm	$\tilde{r}^{[\#1]}$	pseudo residuals
\rmi	$\tilde{r}^{[\#1](i)}$	pseudo residuals
\Rtm	$R_t^{[\#1]}$	terminal-region
\Tm	$T^{[\#1]}$	terminal-region
\ctm	$c_t^{[\#1]}$	mean, terminal-regions
\ctmh	$\hat{c}_t^{[\#1]}$	mean, terminal-regions with hat
\ctmt	$\tilde{c}_t^{[\#1]}$	mean, terminal-regions

<code>\Lp</code>	L'	
<code>\Ldp</code>	L''	
<code>\Lpleft</code>	L'_{left}	
<code>\ts</code>	$\boldsymbol{\theta}^*$	theta*
<code>\bljt</code>	$b^{[j]}(\boldsymbol{x}, \boldsymbol{\theta})$	BL j with theta
<code>\bljts</code>	$b^{[j]}(\boldsymbol{x}, \boldsymbol{\theta}^*)$	BL j with theta*

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ml-eval

Macro	Notation	Comment
<code>\ntest</code>	n_{test}	size of the test set
<code>\ntrain</code>	n_{train}	size of the train set
<code>\ntesti</code>	$n_{\text{test},\#1}$	size of the i-th test set
<code>\ntraini</code>	$n_{\text{train},\#1}$	size of the i-th train set
<code>\Jtrain</code>	J_{train}	index vector train data
<code>\Jtest</code>	J_{test}	index vector test data
<code>\Jtraini</code>	$J_{\text{train},\#1}$	index vector i-th train dataset
<code>\Jtesti</code>	$J_{\text{test},\#1}$	index vector i-th test dataset
<code>\Dtraini</code>	$\mathcal{D}_{\text{train},\#1}$	D_train,i, i-th training set
<code>\Dtesti</code>	$\mathcal{D}_{\text{test},\#1}$	D_test,i, i-th test set
<code>\JSpace</code>	$\{1, \dots, n\}^{\#1}$	space of train indices of size n_train
<code>\JtrainSpace</code>	$\{1, \dots, n\}^{n_{\text{train}}}$	space of train indices of size n_train
<code>\JtestSpace</code>	$\{1, \dots, n\}^{n_{\text{test}}}$	space of train indices of size n_test
<code>\yJ</code>	$\mathbf{y}_{\#1}$	output vector associated to index J
<code>\yJDef</code>	$\left(y^{(J^{(1)})}, \dots, y^{(J^{(m)})}\right)$	def of the output vector associated to index J
<code>\JJ</code>	$\mathcal{J}$	cali-J, set of all splits
<code>\JJset</code>	$((J_{\text{train},1}, J_{\text{test},1}), \dots, (J_{\text{train},B}, J_{\text{test},B}))$	(Jtrain_1,Jtest_1) ...(Jtrain_B,Jtest_B)
<code>\Itrainlam</code>	$\mathcal{I}(\mathcal{D}_{\text{train}}, \boldsymbol{\lambda})$	
<code>\GE</code>	GE	GE
<code>\GEh</code>	$\widehat{\text{GE}}$	GE-hat
<code>\GEfull</code>	$\text{GE}(\mathcal{I}, \boldsymbol{\lambda}, \#1, \rho)$	GE full
<code>\GEholdout</code>	$\widehat{\text{GE}}_{J_{\text{train}}, J_{\text{test}}}(\mathcal{I}, \boldsymbol{\lambda}, J_{\text{train}} , \rho)$	GE hat holdout
<code>\GEholdouti</code>	$\widehat{\text{GE}}_{J_{\text{train},\#1}, J_{\text{test},\#1}}(\mathcal{I}, \boldsymbol{\lambda}, J_{\text{train},\#1} , \rho)$	GE hat holdout i-th set
<code>\GEhlam</code>	$\widehat{\text{GE}}(\boldsymbol{\lambda})$	GE-hat(lam)
<code>\GEhlamsubIJrho</code>	$\widehat{\text{GE}}_{\mathcal{I}, \mathcal{J}, \rho}(\boldsymbol{\lambda})$	GE-hat_I,J,rho(lam)
<code>\GEhresa</code>	$\widehat{\text{GE}}(\mathcal{I}, \mathcal{J}, \rho, \boldsymbol{\lambda})$	GE-hat_I,J,rho(lam)
<code>\GERhoDef</code>	$\lim_{n_{\text{test}} \rightarrow \infty} \mathbb{E}_{\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \sim \mathbb{P}_{xy}} [\rho(\mathbf{y}_{J_{\text{test}}}, \mathbf{F}_{J_{\text{test}}, \mathcal{I}(\mathcal{D}_{\text{train}}, \boldsymbol{\lambda})})]$	GE formal def
<code>\agr</code>	agr	aggregate function
<code>\GEf</code>	$\text{GE}(\hat{f})$	GE of a fitted model
<code>\GEfh</code>	$\widehat{\text{GE}}(\hat{f})$	GEh of a fitted model
<code>\GEfL</code>	$\text{GE}(\hat{f}, L)$	GE of a fitted model wrt loss L
<code>\Lyfhx</code>	$L(y, \hat{f}(\mathbf{x}))$	pointwise loss of fitted model
<code>\GEnf</code>	$GE_n(\hat{f}_{\#1})$	GE of a fitted model
<code>\GEind</code>	$GE_n(\mathcal{I}_{L,O})$	GE of inducer
<code>\GED</code>	$\text{GE}_{\mathcal{D}}$	GE indexed with data
<code>\EGEn</code>	EGE_n	expected GE
<code>\EDn</code>	$\mathbb{E}_{ D =n}$	expectation wrt data of size n
<code>\rhoL</code>	ρ_L	perf. measure derived from pointwise loss
<code>\F</code>	$\mathbf{F}$	matrix of prediction scores

<code>\Fi</code>	$\mathbf{F}^{(\#1)}$	i-th row vector of the predscore mat
<code>\FJ</code>	$\mathbf{F}_{\#1}$	predscore mat idxvec J
<code>\FJf</code>	$\mathbf{F}_{J,f}$	predscore mat idxvec J and model f
<code>\FJtestfh</code>	$\mathbf{F}_{J_{\text{test}},\hat{f}}$	predscore mat idxvec Jtest and model f hat
<code>\FJtestftrain</code>	$\mathbf{F}_{J_{\text{test}},\mathcal{I}(\mathcal{D}_{\text{train}},\boldsymbol{\lambda})}$	predscore mat idxvec Jtest and model f
<code>\FJtestftraini</code>	$\mathbf{F}_{J_{\text{test}},\#1,\mathcal{I}(\mathcal{D}_{\text{train}},\#1,\boldsymbol{\lambda})}$	predscore mat i-th idxvec Jtest and model f
<code>\FJfDef</code>	$\left(f(\mathbf{x}^{(J^{(1)})}), \dots, f(\mathbf{x}^{(J^{(m)})})\right)$	def of predscore mat idxvec J and model f
<code>\preimageRho</code>	$\bigcup_{m \in \mathbb{N}} (\mathcal{Y}^m \times \mathbb{R}^{m \times g})$	Set of all datasets times HP space
<code>\np</code>	n_+	no. of positive instances
<code>\nn</code>	n_-	no. of negative instances
<code>\rn</code>	π_-	proportion negative instances
<code>\rp</code>	π_+	proportion positive instances
<code>\tp</code>	$\#TP$	true pos
<code>\fap</code>	$\#FP$	false pos (fp taken for partial derivs)
<code>\tn</code>	$\#TN$	true neg
<code>\fan</code>	$\#FN$	false neg

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ml-feature-sel

Macro	Notation	Comment
<code>\xjNull</code>	x_{j_0}	
<code>\xjEins</code>	x_{j_1}	
<code>\xl</code>	$\mathbf{x}_l$	
<code>\pushcode</code>		

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ml-gp

Macro	Notation	Comment
<code>\fvec</code>	$[f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(n)})]$	$\{f(x_1), \dots, f(x_n)\}$
<code>\fv</code>	$\mathbf{f}$	bold f, function vector
<code>\mv</code>	$\mathbf{m}$	bold m, GP mean vector
<code>\kv</code>	$\mathbf{k}$	bold k, kernel mat partition
<code>\kcc</code>	$k(\cdot, \cdot)$	$k(\cdot, \cdot)$, kernel for arbitrary inputs
<code>\kxij</code>	$k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$	$k(x_i, x_j)$, cov of x_i, x_j
<code>\Kmat</code>	$\mathbf{K}$	K, kernel mat
<code>\Nmk</code>	$\mathcal{N}(\mathbf{m}, \mathbf{K})$	$n(m, K)$, Gaussian w/ mean vec, cov mat
<code>\Nzk</code>	$\mathcal{N}(\mathbf{0}, \mathbf{K})$	$n(0, K)$, zero-mean Gaussian
<code>\GPmk</code>	$\mathcal{GP}(m(\cdot), k(\cdot, \cdot))$	$GP(m(\cdot), k(\cdot, \cdot))$, GP definition
<code>\GPzk</code>	$\mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$	$GP(0, k(\cdot, \cdot))$, zero-mean GP
<code>\Xsubset</code>	$\mathbf{X}$	bold X, finite subset from xspace
<code>\fX</code>	$f(\mathbf{X})$	$f(X)$, function vector of finite subset
<code>\kXX</code>	$k(\mathbf{X}, \mathbf{X})$	$k(X, X)$, cov for finite subset
<code>\mX</code>	$m(\mathbf{X})$	$m(X)$, mean for finite subset
<code>\ls</code>	ℓ	length-scale
<code>\xxtnorm</code>	$\ \mathbf{x} - \tilde{\mathbf{x}}\ $	$\ x - x_{\text{tilde}}\ $
<code>\xs</code>	$\mathbf{x}_*$	x_* , test obs features
<code>\ys</code>	$\mathbf{y}_*$	y_* , test obs target
<code>\fs</code>	$\mathbf{f}_*$	f_* , test obs fun vector
<code>\Xs</code>	$\mathbf{X}_*$	X_* , test design matrix
<code>\ks</code>	$\mathbf{k}_*$	k_* , cov vec of new obs with x
<code>\kss</code>	$\mathbf{k}_{**}$	k_{**} , cov vec of new obs
<code>\Ks</code>	$\mathbf{K}_*$	K_* , cov mat of new obs with x
<code>\Kss</code>	$\mathbf{K}_{**}$	K_{**} , cov mat of new obs
<code>\Kinv</code>	$\mathbf{K}^{-1}$	K^{-1} , inverse cov mat
<code>\Ky</code>	$\mathbf{K}_y$	K_y , cov mat of y

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ml-hpo

Macro	Notation	Comment
\Ilam	$\mathcal{I}_{\lambda}$	inducer with HP
\LamS	$\tilde{\Lambda}$	search space
\lami	$\lambda^{(\#1)}$	lambda i
\clam	$c(\lambda)$	c(lambda)
\clamh	$c(\hat{\lambda})$	c(lambda-hat)
\lams	λ^*	theoretical min of c
\lamh	$\hat{\lambda}$	returned lambda of HPO
\lamp	λ^+	proposed lambda
\clamp	$c(\lambda^+)$	c of proposed lambda
\archive	$\mathcal{A}$	archive
\archivet	$\mathcal{A}^{[\#1]}$	archive at time step t
\tuner	$\mathcal{T}$	tuner
\tunerfull	$\mathcal{T}_{\mathcal{I}, \tilde{\Lambda}, \rho, \mathcal{J}}$	tuner with inducer, search space, perf measure, resampling strategy
\chlam	$\hat{c}(\lambda)$	post mean of SM
\shlam	$\hat{\sigma}(\lambda)$	post sd of SM
\vhlam	$\hat{\sigma}^2(\lambda)$	post var of SM
\ulam	$u(\lambda)$	acquisition function
\lambdaopt	λ^*	minimum of the black box function Psi
\metadata	$\{(\lambda^{(i)}, \Psi^{[i]})\}$	metadata for the Gaussian process
\lamvec	$(\lambda^{[1]}, \dots, \lambda^{[m_{\text{init}}]})$	vector of different inputs
\minit	m_{init}	size of the initial design
\lambu	λ_{budget}	single lambda_budget component HP
\lamfid	λ_{fid}	single lambda_fidelity
\lamfidl	$\lambda_{\text{fid}}^{\text{low}}$	single lambda_fidelity lower
\lamfidu	$\lambda_{\text{fid}}^{\text{upp}}$	single lambda_fidelity upper
\etahb	η_{HB}	HB multiplier eta

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ml-infotheory

Macro	Notation	Comment
<code>\entx</code>	$-\sum_{x \in \mathcal{X}} p(x) \cdot \log p(x)$	entropy of x
<code>\dentx</code>	$-\int_{\mathcal{X}} f(x) \cdot \log f(x) dx$	diff entropy of x
<code>\jentyx</code>	$-\sum_{x \in \mathcal{X}} p(x, y) \cdot \log p(x, y)$	joint entropy of x, y
<code>\jdentyx</code>	$-\int_{\mathcal{X}, \mathcal{Y}} f(x, y) \cdot \log f(x, y) dx dy$	joint diff entropy of x, y
<code>\centyx</code>	$-\sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y x) \cdot \log p(y x)$	cond entropy y x
<code>\cdentyx</code>	$-\int_{\mathcal{X}, \mathcal{Y}} f(x, y) \cdot \log f(y x) dx dy$	cond diff entropy y x
<code>\xentpq</code>	$-\sum_{x \in \mathcal{X}} p(x) \cdot \log q(x)$	cross-entropy of p, q
<code>\kl dpq</code>	$D_{KL}(p q)$	KLD between p and q
<code>\kl dpqt</code>	$D_{KL}(p q_{\theta})$	KLD divergence between p and parameterized q
<code>\explogpq</code>	$\mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right]$	expected LLR of p, q (def KLD)
<code>\sumlogpq</code>	$\sum_{x \in \mathcal{X}} p(x) \cdot \log \frac{p(x)}{q(x)}$	expected LLR of p, q (def KLD)

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ml-interpretable

Macro	Notation	Comment
<code>\pert</code>	$\tilde{\#1}^{\#2 \#3}$	command to express that for #1 the subset #2 was perturbed given subset #3
<code>\fj</code>	f_j	marginal function f_j , depending on feature j
<code>\fnj</code>	f_{-j}	marginal function f_{-j} , depending on all features but j
<code>\fS</code>	f_S	marginal function f_S depending on feature set S
<code>\fC</code>	f_C	marginal function f_C depending on feature set C
<code>\fhj</code>	$\hat{f}_j$	marginal function fh_j , depending on feature j
<code>\fhmj</code>	$\hat{f}_{-j}$	marginal function fh_{-j} , depending on all features but j
<code>\fhS</code>	$\hat{f}_S$	marginal function fh_S depending on feature set S
<code>\fhC</code>	$\hat{f}_C$	marginal function fh_C depending on feature set C
<code>\XSmat</code>	$\mathbf{X}_S$	Design matrix subset
<code>\XCmat</code>	$\mathbf{X}_C$	Design matrix subset
<code>\Xnj</code>	$\mathbf{X}_{-j}$	Design matrix subset $-j = \{1, \dots, j-1, j+1, \dots, p\}$
<code>\fhice</code>	$\hat{f}_{\#1, ICE}$	ICE function
<code>\Scupj</code>	$S \cup \{j\}$	coalition S but without player j
<code>\Scupk</code>	$S \cup \{k\}$	coalition S but without player k
<code>\SsubP</code>	$S \subseteq P$	coalition S subset of P
<code>\SsubPnoj</code>	$S \subseteq P \setminus \{j\}$	coalition S subset of P without player j
<code>\SsubPnojk</code>	$S \subseteq P \setminus \{j, k\}$	coalition S subset of P without player k
<code>\phiij</code>	$\hat{\phi}_j^{(i)}$	Shapley value for feature j and observation i
<code>\Gspace</code>	$\mathcal{G}$	Hypothesis space for surrogate model
<code>\neigh</code>	$\phi_{\mathbf{x}}$	Proximity measure
<code>\zv</code>	$\mathbf{z}$	Sampled datapoints for surrogate
<code>\Gower</code>	d_G	Gower distance

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ml-mbo

Macro	Notation	Comment
<code>\xvsi</code>	$\mathbf{x}^{[\#1]}$	x at iteration i
<code>\ysi</code>	$y^{[\#1]}$	y at iteration i
<code>\Dt</code>	$\mathcal{D}^{[\#1]}$	archive at iteration t
<code>\Dts</code>	$\mathcal{D}^{[t]} = \{(\mathbf{x}^{[i]}, y^{[i]})\}_{i=1, \dots, t}$	archive at iteration t fully
<code>\fh</code>	$\hat{s}$	surrogate mean
<code>\sh</code>	$\hat{s}$	surrogate se
<code>\fmin</code>	$f_{\min}$	current best

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ml-multitarget

Macro	Notation	Comment
<code>\Tspace</code>	$\mathcal{T}$	
<code>\tv</code>	$\mathbf{t}$	
<code>\tim</code>	$\mathbf{t}_m^{(i)}$	
<code>\yim</code>	$y_m^{(i)}$	

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ml-nn

Macro	Notation	Comment
\neurons	$z_1, \dots, z_M$	vector of neurons
\hidz	$\mathbf{z}$	vector of hidden activations
\biasb	$\mathbf{b}$	bias vector
\biasc	c	bias in output
\wtw	$\mathbf{w}$	weight vector (general)
\Wmat	$\mathbf{W}$	weight vector (general)
\wtu	$\mathbf{u}$	weight vector of output neuron
\Oreg	$R_{reg}(\theta X, y)$	regularized objective function
\Ounreg	$R_{emp}(\theta X, y)$	unconstrained objective function
\Pen	$\Omega(\theta)$	penalty
\Oregweight	$R_{reg}(w X, y)$	regularized objective function with weight
\Oweight	$R_{emp}(w X, y)$	unconstrained objective function with weight
\Oweighti	$R_{emp}(w_i X, y)$	unconstrained objective function with weight w_i
\Oweightopt	$J(w^* X, y)$	unconstrained objective function with optimal weight
\Opt	$\hat{J}(\theta X, y)$	optimal objective function
\Odropout	$J(\theta, \mu X, y)$	dropout objective function
\Loss	$L(y, f(\mathbf{x}, \boldsymbol{\theta}))$	
\Lmomentumnest	$L(y^{(i)}, f(x^{(i)}, \boldsymbol{\theta} + \varphi \boldsymbol{\nu}))$	momentum risk
\Lmomentumtilde	$L(y^{(i)}, f(x^{(i)}, \tilde{\boldsymbol{\theta}}))$	Nesterov momentum risk
\Lmomentum	$L(y^{(i)}, f(x^{(i)}, \boldsymbol{\theta}))$	
\Hess	$\mathbf{H}$	
\nub	$\boldsymbol{\nu}$	
\uauto	$L(x, g(f(x)))$	undercomplete autoencoder objective function
\dauto	$L(x, g(f(\tilde{x})))$	denoising autoencoder objective function
\deltab	$\boldsymbol{\delta}$	
\Lossdeltai	$L(y^{(i)}, f(\mathbf{x}^{(i)} + \boldsymbol{\delta} \boldsymbol{\theta}))$	
\Lossdelta	$L(y, f(\mathbf{x} + \boldsymbol{\delta} \boldsymbol{\theta}))$	

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ml-online

Macro	Notation	Comment
<code>\Aspace</code>	$\mathcal{A}$	
<code>\norm</code>	$ \#1 _2$	
<code>\llin</code>	L^{lin}	
<code>\lzeroone</code>	L^{0-1}	
<code>\lhinge</code>	L^{hinge}	
<code>\lexphinge</code>	$\widetilde{L^{\text{hinge}}}$	
<code>\lconv</code>	L^{conv}	
<code>\FTL</code>	FTL	
<code>\FTRL</code>	FTRL	
<code>\OGD</code>	OGD	
<code>\EWA</code>	EWA	
<code>\REWA</code>	REWA	
<code>\EXPthree</code>	EXP3	
<code>\EXPthreep</code>	EXP3P	
<code>\reg</code>	ψ	
<code>\Algo</code>	Algo	

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ml-regu

Macro	Notation	Comment
<code>\thetas</code>	$\boldsymbol{\theta}^*$	theta star
<code>\thetaridge</code>	$\boldsymbol{\theta}_{\text{ridge}}$	theta (RIDGE)
<code>\thetalasso</code>	$\boldsymbol{\theta}_{\text{LASSO}}$	theta (LASSO)
<code>\thetaols</code>	$\boldsymbol{\theta}_{\text{OLS}}$	theta (RIDGE)

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ml-survival

Macro	Notation	Comment
<code>\Ti</code>	$T^{(\#1)}$	??
<code>\Ci</code>	$C^{(\#1)}$	??
<code>\oi</code>	$o^{(\#1)}$	??
<code>\ti</code>	$t^{(\#1)}$	??
<code>\deltai</code>	$\delta^{(\#1)}$	
<code>\Lxdi</code>	$L(\boldsymbol{\delta}, f(\boldsymbol{x}))$	

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ml-svm

Macro	Notation	Comment
<code>\sv</code>	SV	supportvectors
<code>\sl</code>	ζ	slack variable
<code>\slvec</code>	$(\zeta^{(1)}, \zeta^{(n)})$	slack variable vector
<code>\sli</code>	$\zeta^{(\#1)}$	i-th slack variable
<code>\scptxi</code>	$\langle \boldsymbol{\theta}, \mathbf{x}^{(i)} \rangle$	scalar prodct of theta and xi
<code>\svmhplane</code>	$y^{(i)} (\langle \boldsymbol{\theta}, \mathbf{x}^{(i)} \rangle + \theta_0)$	SVM hyperplane (normalized)
<code>\alphah</code>	$\hat{\alpha}$	alpha-hat (basis fun coefficients)
<code>\alphav</code>	$\boldsymbol{\alpha}$	vector alpha (bold) (basis fun coefficients)
<code>\alphavh</code>	$\hat{\boldsymbol{\alpha}}$	vector alpha-hat (basis fun coefficients)
<code>\dualobj</code>	$\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} \langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle$	min objective in lin svm dual
<code>\HS</code>	Φ	H, hilbertspace
<code>\phix</code>	$\phi(\mathbf{x})$	feature map x
<code>\phixt</code>	$\phi(\tilde{\mathbf{x}})$	feature map x tilde
<code>\kxxt</code>	$k(\mathbf{x}, \tilde{\mathbf{x}})$	kernel fun x, x tilde
<code>\scptxifm</code>	$\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle$	scalar prodct of theta and xi

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ml-trees

Macro	Notation	Comment
<code>\Np</code>	$\mathcal{N}$	(Parent) node N
<code>\Npk</code>	$\mathcal{N}_k$	node N_k
<code>\Nl</code>	$\mathcal{N}_1$	Left node N_1
<code>\Nr</code>	$\mathcal{N}_2$	Right node N_2
<code>\pikN</code>	$\pi_{\#1}^{(\mathcal{N})}$	class probability node N
<code>\pikNh</code>	$\hat{\pi}_{\#1}^{(\mathcal{N})}$	estimated class probability node N
<code>\pikNlh</code>	$\hat{\pi}_{\#1}^{(\mathcal{N}_1)}$	estimated class probability left node
<code>\pikNr</code>	$\hat{\pi}_{\#1}^{(\mathcal{N}_2)}$	estimated class probability right node

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